

École Polytechnique Fédérale de Lausanne

AI Tools: A Digital Activity to Teach K-12 about Machine Learning

by Eglantine Vialaneix



Semester Project Report

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Titre: AI TOOLS | Impact of glass-boxing the Machine Learning pipeline on learning

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Donnée & travail demandé :

Machine learning (ML) becomes increasingly prevalent in everyday technology, understanding its foundational principles is essential for future generations. However, ML systems—often described as "black boxes"—can be challenging to comprehend, presenting a unique challenge when introducing these concepts to novices or students. One way to make ML more approachable is to foreground the stages of an ML pipeline: data collection, data cleaning, feature selection, model training, and evaluation. Prior work suggests that involving students in data preparation can improve AI understanding, but fewer tools have tested whether transparency into model logic provides similar benefits.

Objective: This project evaluates the concept of pipeline transparency by testing how different types of transparency—data preparation and model logic—affect students' understanding of ML.

We use decision trees (DTs) as the target model because their structure is generally considered intuitive and interpretable for students. To investigate these questions, we will develop a custom web-based platform to allow running studies comparing three levels of transparency.

Remarques :

Un plan de travail sera établi et présenté aux assistants avant le vendredi 27 février 2026.

Une présentation intermédiaire (environ 10 minutes de présentation et 10 minutes de discussion) de votre travail aura lieu dans le courant du mois d'avril 2026. Elle a pour objectifs de donner un rapide résumé du travail déjà effectué, de proposer un plan précis pour la suite du projet et d'en discuter les options principales.

Un rapport, comprenant en son début un titre avec les données générales et une photo portrait de l'étudiant, suivi par l'énoncé du travail (présent document), puis d'un résumé d'une page (selon canevas), sera remis le lundi 1 juin 2026 avant 12 heures en format électronique à l'assistant responsable. L'accent sera mis sur les expériences et les résultats obtenus. Le public cible est de type ingénieur EPF sans connaissance pointue du domaine. Une version préliminaire du rapport sera remise à l'assistant le 15 mai 2026. Tous les documents en version informatique, y compris le rapport (aussi en version source), le document de la présentation orale, ainsi que les sources des différents programmes doivent être rendus selon les consignes qui seront données avant la terminaison du projet.

Une défense de 30 minutes (environ 20 minutes de présentation et démonstration, plus 10 minutes de réponses aux questions) aura lieu dans la période du 1 au 12 juin 2026.

Le professeur responsable:

L'assistant responsable:

Signature : 	Signature : 
Francesco Mondada	Valentina Ferraioli

Lausanne, le 16 février 2026

Now, you've lived the full experience Pierre Dillenbourg told you about:

"You make a cool technology, and then you cry."

— Valentina Ferraioli

Abstract

Artificial intelligence (AI) is increasingly present in children's everyday lives, but it is often encountered as a black box: children can see what goes in and what comes out, without understanding what happens in between. This project explores whether making the machine learning pipeline more transparent can support K-12 students' understanding of AI.

To investigate this question, we designed and implemented *AI Tools*, an interactive web platform introducing teenagers aged 13 to 14 to four steps of a supervised machine learning pipeline: data labelling, feature selection, model training, and evaluation. The activity is framed as a dinosaur classification mission, where students train a model to classify dinosaurs as herbivores or carnivores. The platform was designed with three experimental conditions, varying how much of the pipeline is glass-boxed or black-boxed.

A first pilot experiment was conducted with ten students in 10^{ème}. Due to classroom and data collection limitations, the pilot did not allow us to measure learning gains or answer the research question conclusively. However, it showed that the platform works technically and can be implemented in classroom setups.

The pilot also highlighted several necessary improvements, especially regarding classroom orchestration, teacher involvement, scaffolding, and the design of the black-box condition. Overall, this project resulted in a promising first version of *AI Tools*, but further iterations are needed before conducting a more reliable experiment on the effect of opening the black box in K-12 machine learning education.

Contents

Abstract	1
1 Introduction	4
2 Background	6
2.1 Machine Learning	6
2.1.1 ML Pipeline	7
2.1.2 Decision Trees	7
2.2 ML in K-12 Education	8
2.2.1 Glass Box Vs Black Box	9
3 Study Design	10
3.1 Learning Goals and Target Group	10
3.2 Scenario Overview	11
3.3 Questionnaires and Data Collection	12
4 Implementation	13
4.1 Technical details	13
4.2 First Iteration	14
4.3 AI Tools	15
4.3.1 Data Labelling	15
4.3.2 Feature Selection	17
4.3.3 Model Training	20
4.3.4 Evaluation	23
5 Pilot Experiment	25
5.1 Discussion	28
5.1.1 Next Steps	29
6 Conclusion	31
Disclosure	33

Acknowledgments	34
Bibliography	35
A Questionnaires	36
A.1 Pre-Test	37
A.2 Post-Test	38
B Activities Screenshots	41
B.1 Data Labelling	42
B.2 Model Training	45
B.3 Evaluation	46
C Pilot Experiment Data	48

Chapter 1

Introduction

Artificial intelligence (AI) is increasingly present in children's everyday lives. They encounter it through chatbots, recommendation systems, image generation tools, search engines, or automated decision systems. As Touretzky et al., 2019 argue, this growing presence makes it important to define what young students should know about AI in order to become informed citizens. However, these technologies are often experienced only through their inputs and outputs: children ask a question, upload an image, click on a button, and receive an answer. What happens in between usually remains hidden.

This opacity is often described through the metaphor of the *black box*. In many AI systems, users can observe what goes in and what comes out, but not how the system transforms the input into an output. For educational purposes, this can be problematic. If students only interact with finished AI tools, they may develop an incomplete understanding of machine learning and imagine that the system is simply "intelligent" by itself, without seeing the role of data, features, training, and evaluation. This is also highlighted by Marques et al., who show that many school-level ML activities make machine learning accessible, but often black-box some of its underlying processes Marques et al., 2020.

Several works in K-12 machine learning education, such as Hitron et al., 2019 and Morales-Navarro and Kafai, 2024, therefore argue for more transparent, or *glass-box*, approaches. Instead of hiding the internal mechanisms of the model, these approaches try to make parts of the machine learning pipeline visible and manipulable by students. Prior research suggests that uncovering the black box can support students' understanding of machine learning concepts, especially when they can actively engage with data preparation, model behaviour, and evaluation (see Chapter 2 below for more details).

This semester project follows this direction. Its goal was to design and implement an interactive platform to teach young students the main steps of a supervised machine learning pipeline. Building

on Ferraioli and Lehner, 2026 and Lehner and Landman, 2025 previous works, the activity focuses on a simple classification task: students are placed in the role of archaeologists who need to train an AI model to classify dinosaurs as herbivores or carnivores. Through this scenario, they are introduced to four core steps of the pipeline: data labelling, feature selection, model training, and evaluation.

The central research question this project attempts to address is the following:

Does opening the black box of the machine learning pipeline support students' understanding of machine learning?

To investigate this question, the platform was designed with three experimental conditions. In the white-box condition, all steps are visible and interactive. In the grey-box condition, data preparation is visible, but model training is hidden. In the black-box condition, most of the pipeline is hidden, and students mainly observe the final predictions and evaluate them.

This report presents the design, implementation, and first pilot test of the platform. Chapter 2 introduces the machine learning concepts needed for the project and the distinction between black-box and glass-box approaches. Chapter 3 presents the study design, including the learning goals, scenario, experimental conditions, and data collection. Chapter 4 describes the implementation of the platform and each activity. Chapter 5 reports the results of the pilot experiment and discusses the main limitations and future improvements.

Chapter 2

Background

2.1 Machine Learning

Machine learning (ML) is a subfield of Artificial intelligence where machines learn patterns from data, and take decisions on their own, instead of being explicitly programmed. ML is often used for tasks like regression, classification or clustering. ML is categorized in three main paradigms:

- **Supervised Learning**, where the model learns from labelled data. Common tasks include Classification and Regression.
- **Unsupervised Learning**, where the model learns relationships/patterns among data. Common tasks include Clustering (finding unknown groupings among the data) and Dimensionality Reduction (reducing the data to fewer features while preserving all relevant information)
- **Reinforcement Learning**: where an agent learns from interaction with its environment and by getting feedback in the form of rewards or penalties. It's usually used for sequential decision making tasks, like in games or robotics.

In this project the focus is on **supervised learning**, and more specifically on a **classification** task, considering its educational purpose. Its aim is, given a set of labelled examples, to train a model to assign the correct category to new, unseen inputs. To reach this goal ML projects follow a structured workflow: first, raw data must be prepared, then a model must be built and configured, and finally its performance must be assessed. The exact steps depend on the learning paradigm. In supervised learning, this workflow takes the form of a well-defined sequence of steps known as the **ML pipeline**, described in the following section.

2.1.1 ML Pipeline

Machine Learning (ML) pipeline is a series of steps necessary to build and deploy a ML model. It allows to get from raw data to predictions. The main steps it goes through are the following:

1. **Data Labelling** is the first step. It consists of annotating each example at our disposal with the correct output category. These annotations, called labels, define what the model will try to predict.
2. **Features Selection** is the second step. Indeed, data are often described through a large set of features (e.g., a dinosaur can be described by its length, weight, diet, period, or taxonomy). However, not all of them are equally useful for a given task. Feature selection is the process of identifying the ones that are more informative for distinguishing the target categories.
3. **Model Training** is the third step. Given the labelled examples and the selected features, the algorithm constructs a decision function that maps inputs to output labels. In this step, the algorithm uses the training data to build internal rules that minimise prediction errors. Different algorithms can be used for this step depending on the complexity of the task, for instance Logistic Regression, Decision Tree, Random Forest, Neural Networks etc. In this project, we use a **Decision Tree**, which will be described in the following section (see Section 2.1.2).
4. **Evaluation**, is the last step. A trained model must be tested on data it has never encountered before, to assess how well it generalises beyond the training examples. Evaluation reveals whether the model has learned meaningful patterns, or has simply memorised the training set. If performance is not satisfactory, it also signals that earlier steps — labelling, feature selection, and/or training — should be revisited.



Figure 2.1: The typical four steps of the Machine Learning Pipeline, here in French: 1) Data Labelling, 2) Features Selection, 3) Model/Algorithm Training and 4) Model/Algorithm Evaluation

2.1.2 Decision Trees

Among the many algorithms available for supervised classification, **decision trees** are particularly well suited to *educational contexts* due to their interpretability and intuitiveness. Indeed, once trained, a decision tree can be read as a series of if-then rules that students can follow and reason about without the need of any particular background. A decision tree classifies inputs by asking a sequence of binary questions about their features, following a branching structure until reaching a leaf node that assigns a class label. Each internal node corresponds to a feature, and each branch corresponds to a possible value or threshold for that feature.

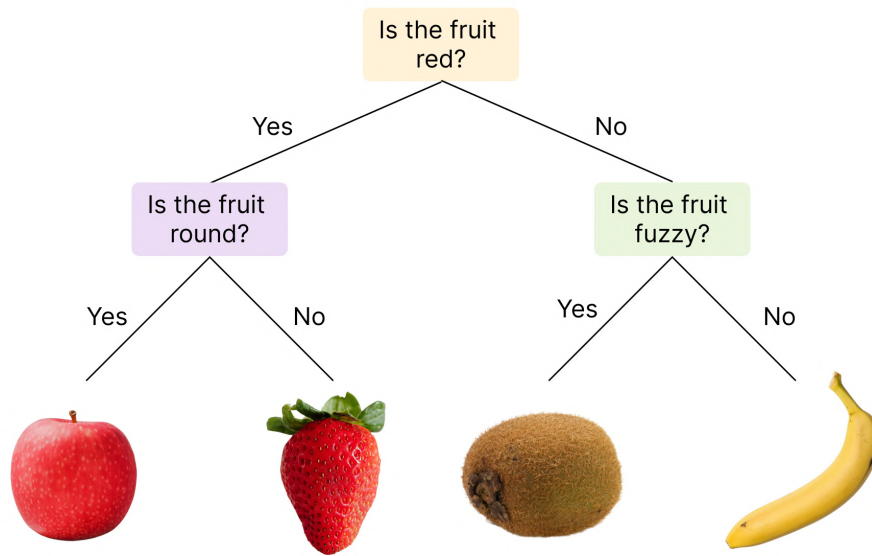


Figure 2.2: Example of Decision Tree. From the root node, several Decisions are successively made based on simple binary questions, to categorize a fruit into the 4 categories, that can be found at the leaf nodes, at the bottom of the tree.

2.2 ML in K-12 Education

To effectively teach machine learning concepts, educational designs must be grounded in recognized competencies. The AI4K12 framework defines the major ideas K-12 students should know about artificial intelligence Touretzky et al., 2019. Aligning with these frameworks, a big amount of work has been done to make ML accessible.

Tools such as Google Teachable Machine make it very easy to train ML models from images, sounds, or poses without coding (Google, n.d.). Code.org's AI for Oceans lets students train a model to distinguish sea creatures from trash and introduces ideas such as training data and bias (Code.org, n.d.). MIT RAISE and Day of AI also provide broader K-12 AI literacy curricula, with activities that introduce how AI works, its risks, and how students can design with it (RAISE, n.d.). These tools are valuable because they make ML accessible and engaging.

However, Marques et al. show that many school-level ML activities focus on accessible processes, such as data management or model use, while model learning and testing are often presented more abstractly (Marques et al., 2020). Hitron et al. showed that children can better understand ML concepts when they interact with uncovered building blocks of ML, especially data labelling and evaluation (Hitron et al., 2019). Extending this perspective, Morales-Navarro and Kafai argue that transparency can be implemented at different levels of the machine learning pipeline: some educational approaches focus on making data practices visible, others expose the learning algorithm

itself, while integrated approaches attempt to combine both (Morales-Navarro and Kafai, 2024). Overall, the literature suggests that accessibility alone is not sufficient for supporting students' understanding of machine learning. An important design consideration is therefore determining which parts of the machine learning pipeline should remain hidden and which should be made visible to learners. This question is commonly discussed through the distinction between black-box and glass-box approaches.

2.2.1 Glass Box Vs Black Box

The ML pipeline described above can be implemented with varying degrees of transparency. In many ML systems, the internal mechanisms are hidden from the user: the system receives inputs and produces outputs without making visible how or why decisions are made. Such systems are often described as *black boxes*. While this opacity can make systems easier to use, it can also be problematic in an educational context. If students only see what enters and what leaves the system, they may understand AI as a tool that simply 'knows' or 'answers', without understanding the role of data, features, training and evaluation.

The opposite approach is often described as *glass-boxing*. A glass-box activity does not only let students use an ML system, but makes some of its internal steps visible, manipulable, and open to reflection. This does not mean that every technical detail must always be shown. As Hmelo and Guzdial explain, black-box scaffolding can be useful when a process is hidden because it is not central to the learning goal, while glass-box scaffolding is needed when the hidden process is precisely what students are expected to understand (Hmelo and Guzdial, n.d.).

In this project, the relevant hidden process is the supervised ML pipeline itself: preparing data, selecting features, training a model, and evaluating its behaviour. It follows the same general direction as Ferraioli and Lehner, who investigate whether transparency across different stages of the machine learning pipeline supports students' understanding of ML concepts in K-12 education (Ferraioli and Lehner, 2026) through an unplugged activity. We adopt a similar idea, but implement it in an interactive digital platform. In this report, we use *black-box* to refer to a step where students can observe the input and output, but not the internal mechanism. We use *glass-box* to refer to a step where students can inspect or act on the mechanism itself. To enable a glass-box approach to the modelling stage, decision trees (described in Section 2.1.2) were chosen as the platform's core algorithm. Their intuitive *if-then* branching structure makes the model training process transparent and easier for students to understand. The central question is therefore not whether every part of ML should always be visible, but whether making selected parts of the pipeline visible can help students understand how an AI model is built, why it may fail, and how earlier choices influence later results.

Chapter 3

Study Design

We designed an online learning activity to teach K-12 about the supervised Machine Learning pipeline. In this chapter we first define the target group and the learning goals of the activity, then describe the scenario students are presented with, the experimental conditions we defined to study the effect of glass-boxing on learning, and finally the data we collected to evaluate learning gains.

3.1 Learning Goals and Target Group

This project targets teenagers aged around 14 years old, which in Vaud corresponds to students in 10^{ème} and 11^{ème}, finishing the *cycle secondaire* of mandatory school. It is around this age that the Plan d'Études Romand (the CIIP portal) expects students to explore and process data in varied ways, to compare algorithms and evaluate their solutions, and to analyse the societal implications of digital technologies.

To address these competencies in the context of ML, we designed a learning activity grounded in the AI4K12's framework from Touretzky et al., 2019, which defines the 5 big ideas of AI students should know about, and the different objectives for each school level. Drawing mostly on the **Big Idea #3: Learning** and its objectives for the 6–8 grade band, we defined the following learning goals.

1. Train and evaluate a classification or prediction model using machine learning on a tabular dataset. (3-A-iii (6-8))
2. Know the **four steps** of the creation of an AI algorithm, namely data labelling, feature selection, model training and evaluation, **their order** and **how they influence each other**, ie that labelling and feature selection directly affect the training and the evaluation. (3-A-iv (9-12))
3. Understand the **purpose of each step**:
 - (a) **Data Labelling**: understand that the data needs to be prepared for the model, especially

that the ground truth needs to be defined, and that is it what the model will try to guess. (3-C-i (6-8))

- (b) **Feature Selection:** understand that the model looks at the characteristics that we provide to guess the label. Carefully choosing what the model looks at helps the model making meaningful choices during the training. (2-A-iv (3-5))
- (c) **Model Training:** understand that the model follows **its own logic** that it applies over and over until it had finished classifying all training data. (3-A-v)
- (d) **Evaluation:** Know that a model is tested on things it has **never seen before** and that its behaviour might be **different than from the training**. Understand that this step is the place to start to **reiterate** on the model afterwards. (3-A-iii)

3.2 Scenario Overview

The whole activity is framed as a dinosaurs **classification** mission in which students are archaeologists who have just discovered some archives about dinosaurs. Their goal is to build an AI model able to help them classify these new dinosaurs as either **carnivores** or **herbivores**.

As introduced in Chapter 2, building such a classifier involves four main stages of the machine learning pipeline. Accordingly, the activity was structured into four phases, each correspond to the four main steps of building their classifier: data labelling, feature selection, model training, and evaluation. These phases are described in section 4.3

Building on this scenario, the central question of this project is whether making the ML pipeline visible supports learning. To investigate this, we designed three experimental conditions that manipulate pipeline transparency while keeping the scenario and overall structure identical for all students. All conditions go through the same flow: a pre-test on paper (see Section 3.3), the four-step online activity (data labelling, feature selection, model training, and evaluation), and a post-test on paper (see Section 3.3). (see Figure 3.1). The three conditions differ in which steps are glass-boxed or black-boxed:

- C1 (or White Box, WB): all four steps are glass-boxed. Students actively perform each step and can observe the internal logic of the pipeline throughout.
- C2 (or Grey Box): data preparation is glass-boxed, but the modelling step is black-boxed. Students prepare the data themselves but do not see the internal construction of the decision tree.
- C3 (or Black Box, BB): both data preparation and modelling are black-boxed. Students receive a pre-labelled dataset and do not interact with the training process. Only the evaluation step remains glass-boxed.

A key design challenge was keeping all conditions engaged and comparable: in particular, black-boxed steps needed to remain meaningful and require similar time on task, even when the internal logic was not shown to students.

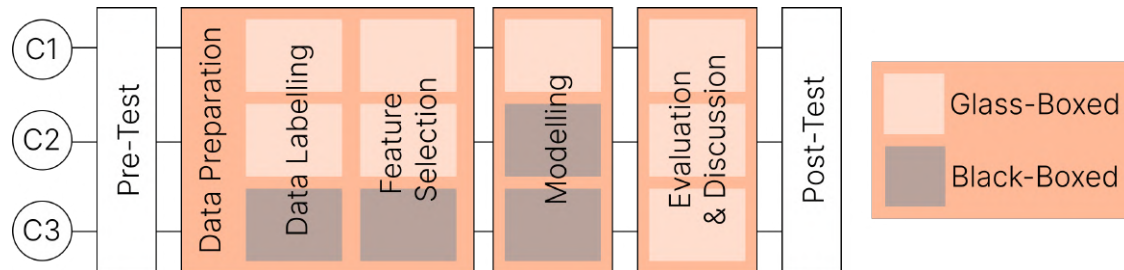


Figure 3.1: Overview of the activity design. Participants are split into three conditions (C1–C3) with increasing Black-Boxing steps. Schema inspired from Ferraioli and Lehner, 2026.

3.3 Questionnaires and Data Collection

We evaluated the learning gains using a qualitative Pre- and Post-Test Evaluation approach. We asked students what they generally knew about AI and the algorithm inside the Black-Box before and after the activity. The pre test contained 3 questions. The post test contained the same first 3 questions and 3 additional ones. Both questionnaires can be found in Chapter A.

We collected data through the app as well to inspect how the students engaged with the activity and to hopefully identify mediators to learning. In general, for each activity (DL, FS, M and E) we collected: - the time that they spent in each activity - the answers that they wrote in each required text field - the number of attempts they make for each required computation. The collected data from the pilot session of the app can be found in the Chapter C, at Table 5.1.

Chapter 4

Implementation

This chapter describes the technical implementation of the AI Tools platform. We first present the technical details of the application — its architecture, the tools used, and the key design decisions. We then describe the first iteration of the platform, highlighting what worked and what needed to change, before presenting the final version of each activity step and how it addresses the learning goals and experimental conditions defined in Chapter 3.

4.1 Technical details

AI Tools was implemented as a web-based learning platform allowing students to progress through the four stages of the supervised machine learning pipeline. The platform adapts its interface according to the assigned experimental condition (white-box, grey-box, or black-box) while preserving the same overall learning scenario. Student interactions are recorded throughout the activity to support later analysis.

The platform is a web application built with a client-server architecture, where the frontend handles all interaction logic and display, and the backend is responsible for computations that cannot run in the browser.

The front-end designs and sketches were made in *Figma*, leveraging the Community *HeroUI* components library, that I sometimes freely adjusted for the app's purposes. I then exported the visual structure and some handcrafted design components into VS Code using the *MCP Figma server*. After this initial conversion, most of the interaction logic was implemented directly in *TypeScript* and JavaScript, built on top of *Next.js*, *React*, and *HeroUI*. This included the activity flow, condition-specific displays, local saving of student answers, table sorting, tutorial hints, warning messages, and transitions between the different activities.

The backend is written in *Python* and exposes a small API that is called by the frontend when

computations are needed. It receives the selected features, the labelled dinosaur data, and the current state of the activity, then returns either Gini scores or model predictions.

For the machine learning computations, a Decision Tree classifier was implemented with *scikit-learn* for the automated tree in BB condition. For the white box, a manually defined Gini impurity computation was implemented using the formula given by Lehner and Landman, 2025. This manual implementation was necessary to make the step-by-step computation visible and inspectable by students, which is central to the glass-boxing approach of the activity.

Deployment and Compatibility. The platform was designed for laptop and desktop use. It was not tested on tablets and is not usable on phones, mainly because the activities require large tables, drag-and-drop-like visual interactions, and side-by-side panels. The app was also tested only on Chromium-based browsers, so compatibility with other browsers is not guaranteed. The application is not deployed online. It is meant to run locally on one computer, with other devices connecting through the same local network. In practice, the frontend and backend are launched separately, and participants access the app through the host computer's IP address.

The application domain content, including the dinosaur information and several teaching materials are inspired by the unplugged activity of Ferraioli and Lehner, 2026. This includes the raw dinosaur table, which we then manually curated and stored in the backend/public data folder, and the dinosaur sheet used during the data labelling activity.

All code of the platform is available on GitHub.

4.2 First Iteration

In this section we briefly mention the first design of the app. This version was tested on three members of the lab, one in each condition. We will highlight the main points and feedback that were pointed out during the pre-pilot session to justify some of the choices that were made to result in the final version of the app.

The first iteration already followed the general structure of the final platform: data labelling, feature selection, model training, and evaluation, and all core interactions were already present. The code for this version can be found on *GitHub* in its dedicated branch.

Three main issues were pointed out:

- The pedagogical instructions were too minimal, which made some transitions between steps not clear for students. At the end of stop steps, the testers stated that they do not know what they learnt from the activity. Steps and instructions needed to be **scaffolded**.
- During model training, students were asked to train and compare three different models, taking different features and amount of data in input. However, this comparison introduced

too many elements at the same time. Students were not familiar with what a model was to begin with and did not know what training meant. This step had to be simplified to the presentation of model training, before being able to analyse and compare between models and performances.

- During the evaluation, students were asked not only to evaluate the performance of the three previous models, but also to use different metrics: accuracy and confusion matrix. The two previous points were stated again there: not enough instructions on how to compute them, and too many new elements to learn at the same time. The evaluation step needed to be better explained and simplified according to the testers.

In general, the final version therefore kept the same global idea as the initial version of the app, but added more explanation and scaffolding, and simplified the model training and evaluation tasks around one main model instead.

4.3 AI Tools

This section presents the final platform and each of its activities in more details. We define here the tasks the students were expected to perform depending on the condition, what we expect them to learn from them by linking them back to our learning goals (see Section 3.1).

4.3.1 Data Labelling

The data labelling activity introduced students to the idea that supervised machine learning starts from labelled examples. The task was to classify a small set of 10 dinosaurs as either herbivores or carnivores. (Figure B.2) The interface, displayed on Figure 4.1, shows a stack of remaining dinosaurs, one main dinosaur card, two target groups, and a note-taking area.

For the glass-boxed groups (C1 and C2), the activity was designed as a simple manipulation task: students read the information sheet of one dinosaur at a time, took notes on a dedicated text field and progressively assigned each dinosaur to one of the two diets. We also made the introduction of wrong labels possible for the student. When this happened, they would get a warning message as displayed in Figure B.3, for them to acknowledge their error and prompt them to pay attention of the impact of their error in the following steps.

In the black-box version, the classification is already annotated on the sheet by a stamp in the corner of the sheet and the card is automatically moved to the correct bucket when moving to the next dinosaur. The students are expected to take a few notes to get familiar with the material and to simply go through the cards. See Figure 4.1.

When all dinosaurs have been assigned, the labels are saved and reused in the next activities.



(a) Glass-Boxed labelling activity (C1 and C2)



(b) Black-Boxed labelling activity (C3)

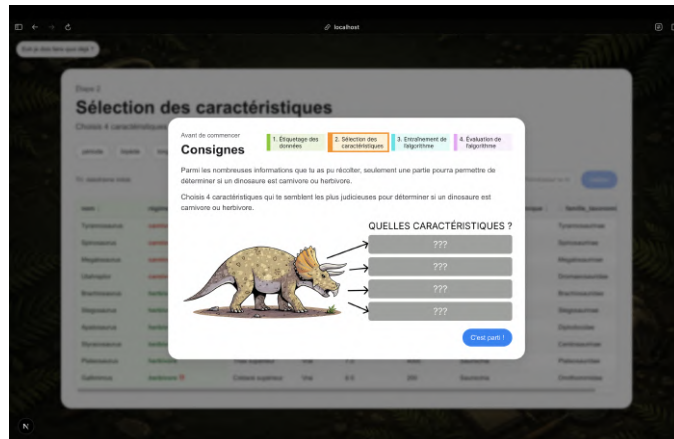
Figure 4.1: Labelling activity page: both conditions see a main card that they are asked to read to identify what can help classify the dinosaurs between carnivore and herbivore. On the top left corner they see the stack of remaining dinosaurs and on the right they see the list of the dinosaurs that they have already classified.

4.3.2 Feature Selection

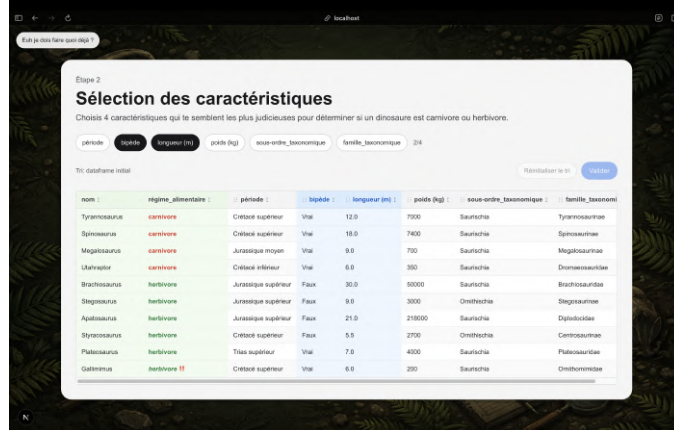
The feature selection activity asked students to inspect the dinosaur data table and reason about the importance of each available characteristics. The table contained the variables period, biped, length, weight, and two taxonomies.

For the Glass-Box conditions (C1 and C2), Students had to identify which features seemed useful for distinguishing carnivores from herbivores. Moreover, the feature selection activity uses the labelled training table, therefore, if a wrong label was made during the previous step, the changed cell is visually marked, so that the link between the activities remains visible (see Figure 4.2, panel (b), bottom line). They are asked to choose the most interesting features according to them, and to provide both a justification and the ordering of the features by their expected importance. The selected features are then stored and used for the following step: model training.

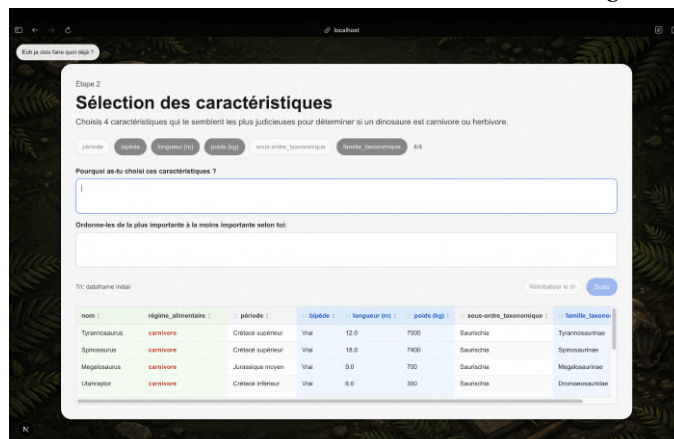
For the C3 group, students are presented an activity that is supposed to "make up" for the time on task their other conditions will spend. To keep them engaged without giving them information of what lies inside the black-box, they were asked to classify each variable as numerical, Boolean, or categorical (see Figure 4.3. Once they've correctly assigned each columns' type, they are all presented the same small search task in the table. Students need to combine three features: sub-order, size, and weight, to find the period of the only dinosaur from the table that is fitting the criteria.



(a) C1 and C2 are presented the feature selection instructions.

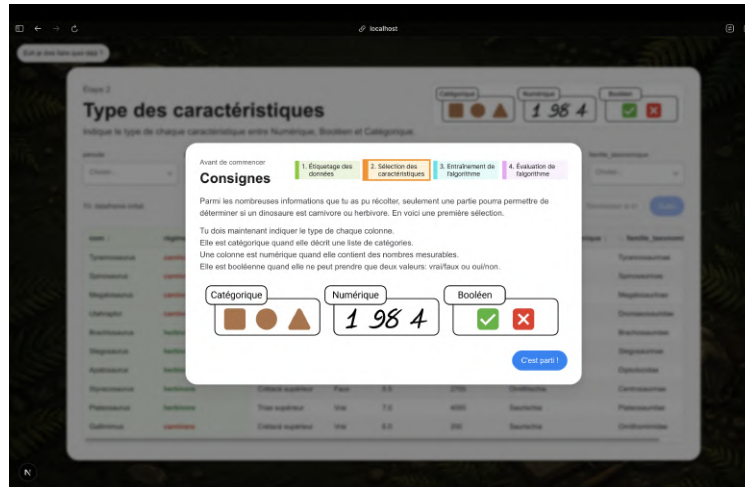


(b) C1 and C2 need to choose the best 4 features according to them.

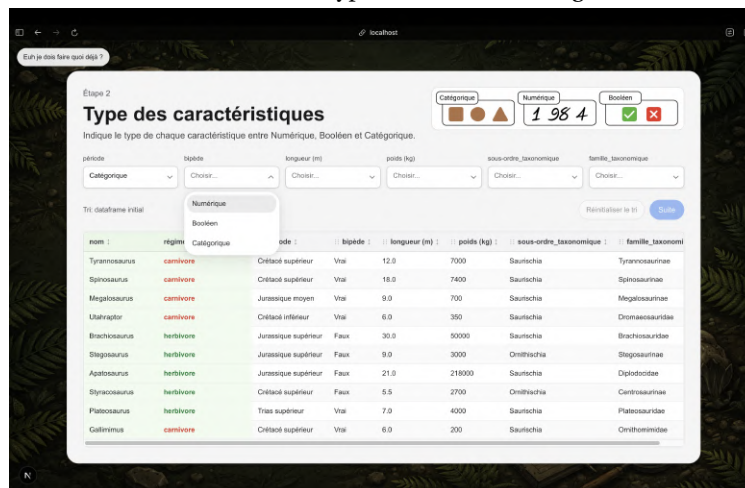


(c) Once they've chosen their features, they are asked to justify their choice and to order their 4 features by importance.

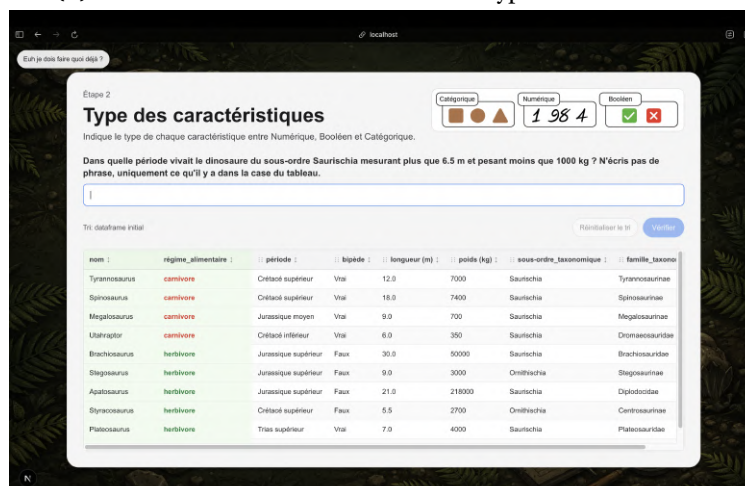
Figure 4.2: Glass-Boxed feature selection activity (C1 and C2). Students are asked to choose the 4 features that they think will be the most important to predict the diet of the dinosaurs.



(a) C3 are presented what are the different types of features: Categorical, Numerical or Boolean.



(b) Students then need to annotate the type of each feature.



(c) Finally, they are asked to perform a search task asking them to navigate between different criteria.

Figure 4.3: Black-Boxed Feature Selection Activity (C3). This activity was designed to keep the students engaged for the same amount of time as the Glass-Boxed groups.

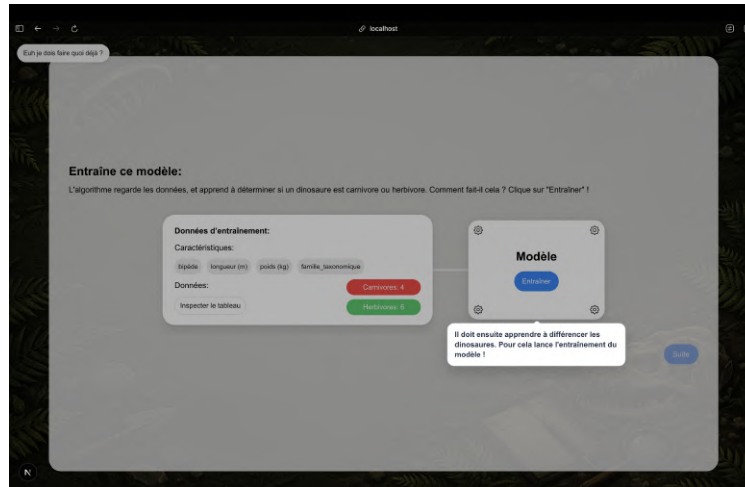
4.3.3 Model Training

The model training activity introduced the central mechanism of the platform: using the prepared data to train a model (See Figure B.4). The students were shown that the algorithm learns from the training set and then produces predictions. All conditions are prompted to click on a simple button to train the model. This was meant to symbolize the autonomy of the model during training, especially for the Black-Boxed groups. (See Figure 4.4) While students in C1 would be redirected to another page showing them what happens inside the model, both C2 and C3 are simply shown the results of the predictions.

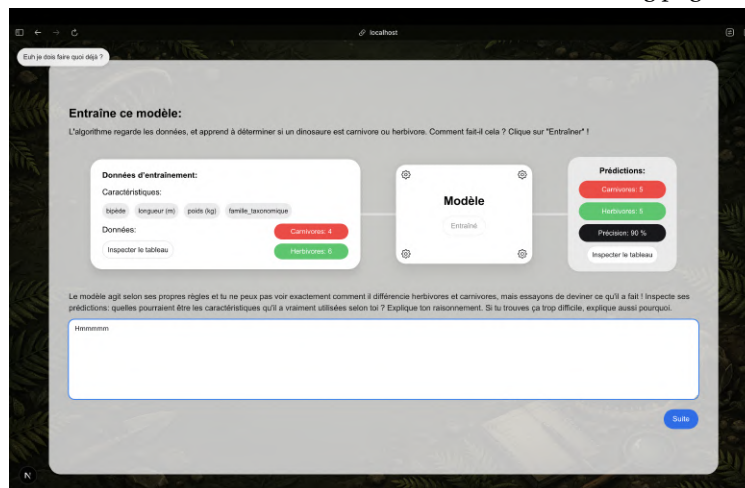
In the white-box version, training the model opens another page where they see the building of the structure of a decision tree (see Figure 4.5). This is where C1 are presented with the concept of Gini Impurity (panel a), and where they are told that the tree follows a simple rule: splitting the data on the feature that creates the purest subsequent groups. At each internal node, they are shown all Gini scores, and can choose to select the best one, or another, or to terminate the building of the tree by defining an early leaf node. This freedom is meant to give them the opportunity to reflect on their actions in the next step: the evaluation.

In the black-boxed conditions C2 and C3, the model is trained through the backend and the student does not see the internal rules. The interface shows the training status for a brief instant, and then a summary of the model predictions on the training data is displayed. Students are asked to inspect the prediction table and to try to infer what the model may have used. They are not expected to understand the rules, but rather to think about what features might have been actually considered during the prediction.

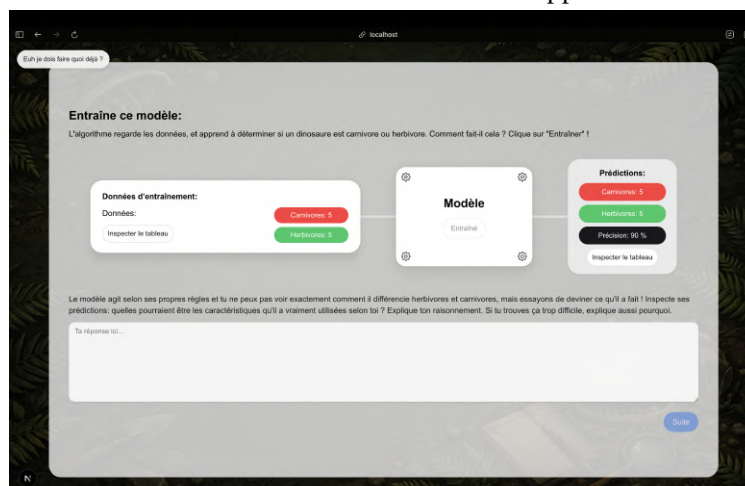
Once the model is trained and that (for C2 and C3 only) all answers have been submitted, the model and its performances are saved to then go to the final step: the evaluation.



(a) C1: can see the selected features and will be directed to the tree building page shown in Figure 4.5.

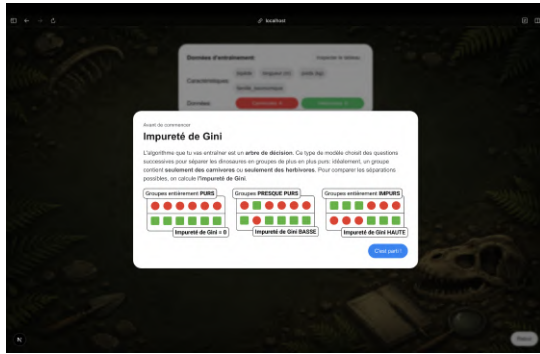


(b) C2: can see their features but need to reflect on what happened inside the Black-Box.

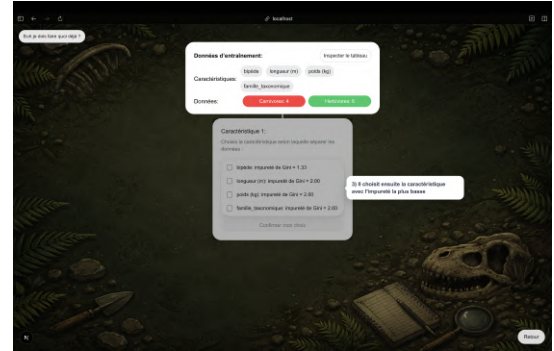


(c) C3: cannot see the features and need to reflect on what happened inside the Black-Box.

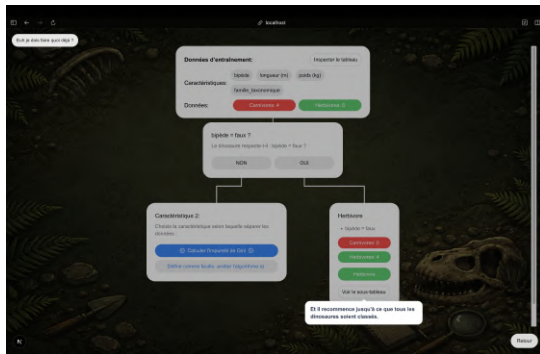
Figure 4.4: Model Training activity compared between each condition: C1 (a), C2 (b) and C3 (c). From the left "input data card", C1 and C2 can see their selected from the past activities. When C1 clicks on "Entraîner", it brings the user to the tree building page shown in Figure 4.5. C2 and C3 do not see the tree, only the prediction card on the right of the model, and are asked to reflect on how its predictions were done.



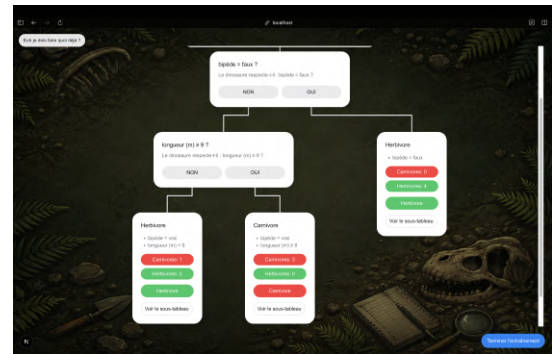
(a) First, the concept of Gini Impurity is presented with a toy example to C1.



(b) Scaffolded instructions show the students what the model uses and how it makes its choice to split the data. The user can then select the features and is free to take the purest split, or not.



(c) C1 students continue to define each split or can terminate the algorithm early.



(d) The final tree is displayed and the number of dinosaurs at the end of each leaf is displayed, allowing students to inspect each leaf group.

Figure 4.5: Interactive building of the Decision Tree. Only visible for the Glass-Boxed Condition C1. The steps encountered by the students are ordered from (a) to (d).

4.3.4 Evaluation

During the evaluation step, all conditions underwent the same activity. Students are told that they will test the model that they trained on dinosaurs that it has never seen before (see Figure B.5. The goal was for them to reflect on whether the model generalizes well to unseen examples.

Students first launch the model prediction, then inspect how many dinosaurs were predicted as carnivores or herbivores. They can open the prediction table to compare the true label with the predicted label. (See Figure 4.6) The only detail that changes, is that C3 do not see the features anymore while C1 and C2 can inspect once more their selected features.

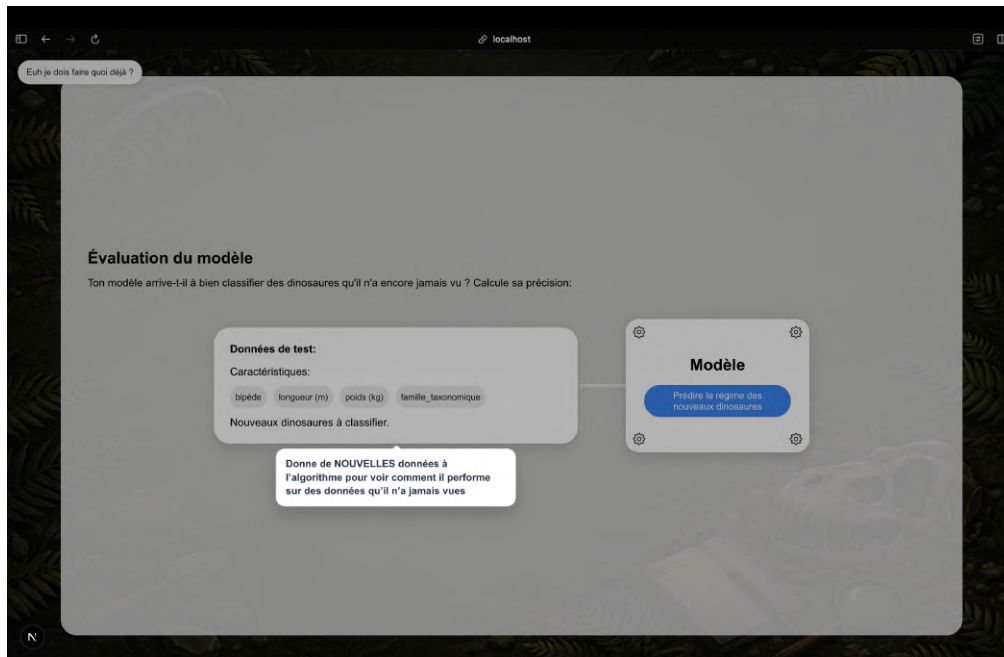
Students are then asked to count first the number of correct predictions, then the number of incorrect ones, and finally to count, or deduce the total number of test dinosaurs. They are then given the formula for the accuracy:

$$\text{accuracy} = \frac{\text{correct predictions}}{\text{total dinosaurs}} \times 100$$

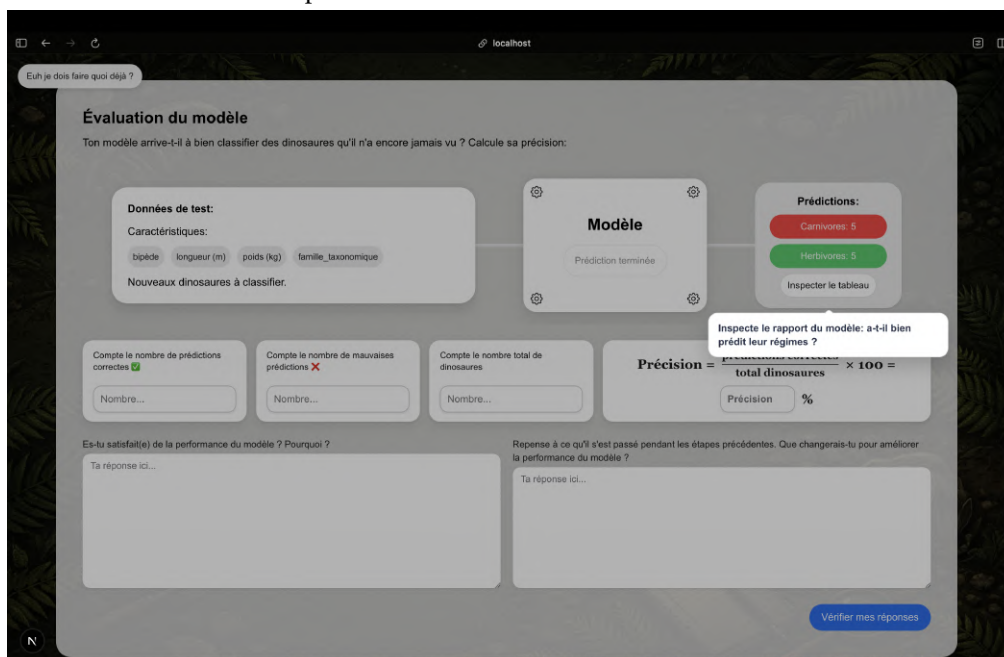
which they can compute using their previous results. The platform checks these answers and highlights incorrect fields if needed.

They were also asked to explain whether they were satisfied with the model's performance and to elaborate on why. Finally, they are asked to think about the previous steps that they went through, and to think about what they would change to improve it.

This last step closes the full machine learning pipeline: students labelled data, selected features, trained a model, tested it on new data, and reflected on its performance. Once they are finished with the activity, students are given the individual post-test.



(a) The "input data" card is shown to highlight the use of unseen dinosaurs. The students need to click on the model to make it predict their diet



(b) Once the model has produced its predictions, the sequence of computation the students needs to perform to compute the accuracy is displayed. They need to open the table and count correct and wrong prediction. (See said tables in Figure B.6)

Figure 4.6: Evaluation activity for all conditions. Students are told that the model that they previously trained will be tested on new data. They will need to assess its performance by computing its accuracy step by step.

Chapter 5

Pilot Experiment

The platform was tested for the first time with its target group during a pilot experiment. The session took place in a classroom of ten male students in 10^{ème} Voie Secondaire Générale, aged between 13 and 14 years old, during their regular informatics class.

Because of technical issues during the pilot, we had to adapt the initial orchestration. Instead of having students work individually or in pairs, they were split into larger groups than expected: three groups of three students and one group of two students, with each group sharing one computer. In total, two groups were assigned to the white-box condition (C1) and two groups to the black-box condition (C3). No group was assigned to the grey-box condition (C2), which limits the comparison between the three planned experimental conditions.

The data collected during the activity are summarized in Table 5.1. Overall, the logs show a high variability in students' engagement across groups and activities. This can be seen in the time spent on each step, but also in the amount and quality of the answers entered in the platform. For instance, during the data labelling activity, some groups took almost no notes and completed the task very quickly, while one group spent much more time and wrote extensive notes on the dinosaur sheets. Similarly, during the evaluation step, all groups had to compute the model accuracy, but the number of attempts varied strongly, from only two attempts in one C3 group to seventeen attempts in one C1 group.

The transcription and translation of the pre- and post-test answers can be found in Table C.1. From the pre-test, we can see that the AI systems students knew about were mostly chatbots or generative tools. Several students described AI as a software or tool that answers questions, uses information, or relies on databases and the internet. This confirms our initial intuition that students of this age are mostly confronted with the input and output of AI systems. They know that something is given to the system and that an answer is produced, but they generally do not have a clear idea of what the algorithm does inside the black box.

Unfortunately, the post-tests do not show clear learning gains. Most of them were handed in almost empty, and students did not spend as much time or focus on them as they did during the pre-test. This makes the comparison between pre- and post-test answers difficult. In addition, the students were very dissipated and interacted a lot with each other, so we cannot guarantee that the questionnaires were answered individually. This is also visible in the answers of both students from group C3 G2, who gave strictly identical answers.

As a consequence, the pilot experiment did not produce exploitable data to answer the initial research question. However, it provided useful feedback on the platform, the classroom orchestration, and the limits of the current version of the activity.

	C1 G1	C1 G2	C3 G1	C3 G2
Data labelling				
Notes taken on dino sheets	None	None	Extensive	None
Incorrect Labels	3	1	[BB]	[BB]
Time on Data Labelling	2	5.7	16.9	2.2
Feature Selection				
Selected Features	Poids, longueur, bipède, période	Poids, longueur, famille taxonomique, période	[BB]	[BB]
Justification	<i>None Given</i>	"Because they are the features with the greatest difference between carnivores and herbivores"	[BB]	[BB]
Feature Importance Order	<i>None Given</i>	Poids, longueur, famille taxonomique, période	[BB]	[BB]
Nb of attempts for Feature types	[GB]	[GB]	1	3
Time on Feature Selection	1.4	5.9	5.3	8.2
Model Training				
Training Accuracy	90	100	100	100
BB Guess of Important features	[GB]	[GB]	"Teeth and Period"	<i>None Given</i>
Time on Model Training	3.3	2.1	1.4	10.2
Evaluation				
Test Accuracy	70	70	30	30
Attempts of accuracy computation	17	16	2	4
Satisfied of performance	Yes	Yes	No	No
Justification	<i>None Given</i>	"Almost all correct"	"Most were wrong"	"It's bad"
Improvements Ideas	"Make it more intuitive and simpler"	<i>None Given</i>	"Give more details"	<i>None Given</i>
Time on Evaluation	10.8	9.6	6.8	12.7

Table 5.1: Summary of each group's answers across the four learning activities. [BB] and [GB] are displayed when the group did not undergo the interaction that was recorded because of its assigned condition.

5.1 Discussion

The goal of this project was to explore whether increasing transparency within the machine learning pipeline could support students' understanding of machine learning concepts. Although the pilot experiment did not produce sufficient evidence to answer this question conclusively, it nevertheless provides several observations that are worth discussing.

Positive Outcomes. First, the platform itself functioned well during the session. All groups were able to go through the activity, and the data were correctly collected. From a technical point of view, this was already an important validation of the implementation.

Second, the activity opened some interesting discussions around trust in AI. Even if students stated that they did not learn much from the activity, and even if we were not able to measure learning gains, some of them still expressed doubts or questions about whether the AI should be trusted. This is encouraging, because one of the broader educational goals of the project is not only to teach the mechanics of machine learning, but also to encourage a more critical attitude towards AI systems.

Third, some answers collected inside the platform provide useful feedback on the activity design. When asked what they would change to improve the model, one C1 group wrote that the activity should be made more intuitive and simpler, while one C3 group asked for more details. This contrast is interesting because it reflects the tension already identified in the design: students need more support to understand the model, but too much information can also overwhelm them.

Finally, the current activity could probably already be tested with an older or more independent target group. This does not mean that the activity should not be used with the current target group, but rather that it needs more adaptation and scaffolding before being used again with students aged 13 to 14.

Limitations. The main limitation of this project is that the pilot did not allow us to measure learning gains in a reliable way. While some students wrote a few meaningful answers in the pre-test, most post-tests were almost empty. We also tried to open a discussion with the classroom after the activity, but most students were not very responsive. Therefore, the qualitative data collected through the questionnaires are too incomplete to support a strong interpretation.

A second limitation concerns classroom management. I genuinely think that I underestimated the importance of the teacher or facilitator in implementing this type of activity in a real classroom. The class was already difficult to manage, and not relying enough on the teacher to help discipline the group added another layer of friction to the experience. This affected the students' concentration, the quality of the answers, and probably their engagement with the activity itself.

A third limitation comes from the design of the activity, especially the model training step. In the current version, I relied too much on the hope that students would reflect *by themselves* on what happened during training. However, if students do not understand the step in the first place, they

may simply not learn anything from it, and I did not take this cases enough in consideration. This was also feedback I received from two members of the lab and from a friend who tried the activity in the black-box condition. In the white-box condition, the visualization of the tree makes the mechanism clearer. However, I neither considered nor implemented an equivalent instructive activity for the black-box, and thus somewhat sabotaged their learning at that step.

Finally, the group format also introduced variability. Some students were active, while others could remain passive behind the computer. Since the activity was not designed with explicit roles, there was no guarantee that each student would engage with all parts of the pipeline. This makes it difficult to interpret both the activity logs and the questionnaire answers at the individual level.

5.1.1 Next Steps

Although the pilot did not provide exploitable data for the research question, it gave me a lot of material to reflect on. If I were to redo the experiment, I would first reconsider the orchestration of the conditions. Instead of having several experimental conditions in the same classroom, I would make the whole class go through the same condition. This would make the logistics easier to handle and would allow students' questions to be answered collectively, without revealing information that should remain hidden for another condition.

I would also involve the teacher more directly. Before the session, I would brief them more carefully about the activity and its goals. Ideally, I would also let them play a stronger role during the introduction and debriefing. Giving power back to the teacher seems essential to me: they know their students better, they already have authority in the classroom, and students are more likely to trust and listen to them than to an external person.

The model training step also needs to be redesigned. I do not have a definitive solution yet, but the current black-box version should probably include more guidance without fully revealing the internal mechanism. This is a difficult balance: students need enough support to make sense of the predictions, but they are already overwhelmed by the amount of text and instructions. One possibility would be to scaffold the training and prediction process dinosaur by dinosaur, in a way similar to AI4Ocean, instead of asking students to reason directly from a full table. I have the impression that the tabular format added too much abstraction to the activity, especially for this age group.

More generally, the activity should either be made fully individual, or redesigned as a real group activity. If students continue to work in groups, each student should have a specific role, and the task should create interdependence between them. This would help ensure that every student participates actively, rather than letting only one student control the computer.

The dinosaur scenario should also be reconsidered carefully. On the one hand, dinosaurs were engaging for several students and testers. The narrative gives students an intuitive understanding of the features and makes the task concrete. On the other hand, it can also be a double-edged sword.

Students who already know a lot about dinosaurs may lose interest in training a model to recognize diet, because they already know that teeth are a very strong indicator. This prior knowledge can bias the activity and reduce the need to reason through the machine learning process.

Finally, I would revise the data labelling task. The current “take notes on the dinosaur sheets” approach led to two extreme behaviours: some students took no notes at all and skipped through the cards very quickly, while others were very diligent and spent a lot of time writing detailed notes. This introduced too much variance between groups. To address this, the activity could return to a more structured format, such as asking students to craft their own table from the dinosaur sheets. This would make the task more constrained and would better prepare the feature selection step.

Overall, the pilot should therefore be understood less as an evaluation of learning gains than as a first usability and classroom test of the platform. It showed that the app can technically support the activity, but also that the pedagogical scaffolding and classroom orchestration need to be strengthened before running a more reliable experiment.

Chapter 6

Conclusion

During this semester, we built a first version of an interactive digital platform, *AI Tools*, to introduce students aged 13 to 14 to four steps of the supervised Machine Learning pipeline: data labelling, feature selection, model training, and evaluation. The activity was designed around a dinosaur classification scenario, where students progressively build an AI model able to classify dinosaurs as herbivores or carnivores.

The main goal of the project was to explore whether opening the black box of machine learning can support learning in K-12 education. To do so, we designed three experimental conditions with different levels of transparency: a white-box condition, a grey-box condition, and a black-box condition. The final platform makes it possible to compare these conditions by changing which parts of the pipeline are visible and interactive for the students.

The platform was tested for the first time with a class of ten students in 10^{ème}. Unfortunately, the pilot did not allow us to determine whether opening the black box supports learning. The post-tests were mostly incomplete, the classroom setting was difficult to control, and the collected data were not sufficient to draw conclusions about learning gains.

However, the pilot was still useful. It showed that the application works technically, that the data collection pipeline functions correctly, and that the activity can generate discussions about trust in AI. It also revealed important limits in the current design, especially regarding classroom orchestration, the role of the teacher, the amount of scaffolding needed, and the difficulty of making the model training step understandable without overwhelming the students.

Future work should therefore focus less on adding new features to the app and more on refining the pedagogical structure around it. In particular, the activity should be better scaffolded, the teacher should be more involved, and the group format should be redesigned to make sure that each student actively participates. The black-box condition also needs to be reconsidered, so that it remains pedagogically meaningful while still preserving the intended difference in transparency.

Overall, this project resulted in a promising first version of *AI Tools*. Even if the pilot did not answer the research question, it provided a clearer understanding of what still needs to be improved before running a more reliable classroom experiment.

Disclosure

I declare having used generative AI during the process of this semester project for the following purposes:

- I used Codex (ChatGPT), GitHub Copilot and ClaudeAI as coding assistants to build the user interface.
- I also used ChatGPT to generate the two background images from the website and enhance the triceratops asset of the instructions.
- Finally, I used ChatGPT to write and to correct my English in parts of the current report.

All prompts and discussions may be requested for inspection.

Finally, to generate this report, I used the IC Overleaf Template, kindly provided by Mathias Payer under Overleaf public license 1.3c.

Acknowledgments

As it is my mother tongue, I wish to express my thanks in French, to make my words as genuine as I'd like them to be received.

J'aimerais remercier du fond du coeur Valentina pour sa contribution à l'écriture de ce rapport, son aide et ses bons conseils tout au long de ce semestre. Non seulement pour sa clairvoyance et son exigence toujours bien placées vis-à-vis de mon projet, mais aussi pour m'avoir offert un précieux coaching de vie personnelle et professionnelle, en ayant toujours répondu à mes diverses questions avec honnêteté et bienveillance.

Un immense merci à Francesco pour m'avoir permis d'entreprendre ce beau projet au sein de son labo, et pour tous les moyens qu'il met à disposition de ses étudiant.e.s. Je me suis sentie à ma place et épaulée dès le début du projet, pour l'avancer autant que j'en apprenais chaque semaine.

Un grand merci à Raimond Trieu pour nous avoir permis de tester AI-Tools dans sa classe, et à ses élèves de nous avoir consacré leur temps, ainsi qu'à Frédérique pour nous avoir mises en contact avec M Trieu.

J'aimerais aussi remercier chaleureusement Johann, Matvey, Adrien, Martin et Sufyan pour avoir accepté de participer aux premiers tests de mon application et pour leurs précieux retours, et à tous.tes les membres du MOBOTS et du LEARN que je n'ai pas encore cité.e.s, qui m'ont gentiment accueillie pendant ce semestre: Jérôme, Kim, Cyril, Manuel, Chenyang.

Merci !

Lausanne, June 2, 2026

Eglantine Vialaneix

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Appendix A

Questionnaires

A.1 Pre-Test

Name/Code: _____

PRE

L'IA, tu connais ? 🤔

Il n'y a pas de bonnes ou mauvaises réponses. On veut juste savoir ce que tu penses avant de commencer !

1

Qu'est-ce que l'intelligence artificielle (IA) pour toi ?

2

Comment penses tu que l'IA devient intelligente ?

3

Qu'est-ce qui se passe quand une IA apprend ?

Complète le schéma: qu'est-ce qui entre ? Qu'est-ce qui se passe au milieu ?
Qu'est-ce qui sort ?

[???] ———> [Algorithmme] ———> [???]

↑

Que fait-il ici ?

A.2 Post-Test

Name/Code: _____

POST

Ce que tu as **retenu** sur l'IA 🙌🧐

Il n'y a pas de bonnes ou mauvaises réponses. On veut juste savoir ce que tu penses après notre activité !

1

Maintenant, qu'est-ce que l'intelligence artificielle pour toi ?

2

Comment l'IA devient-elle intelligente ?

3

Qu'est-ce qui se passe quand une IA apprend ?

Complète le schéma: qu'est-ce qui entre ? Qu'est-ce qui se passe au milieu ?
Qu'est-ce qui sort ?

[???] ———> [Algorithme] ———> [???]

↑

Que fait-il ici ?

Name/Code: _____

POST

Ce que tu as **retenu** sur l'IA 🙌🧐

Il n'y a pas de bonnes ou mauvaises réponses. On veut juste savoir ce que tu penses après notre activité !

4

Ton IA s'est trompée sur un dinosaure : elle a dit carnivore mais c'était un herbivore. D'où pourrait venir cette erreur selon toi ?

5

Ces 4 étapes s'influencent-elles les unes les autres ? Dessine des flèches pour montrer les liens, et écris un mot sur chaque flèche pour expliquer.

Etiquetage

Sélection des
Caractéristiques

Entrainement

Evaluation

Name/Code: _____

6

Un musée aimerait utiliser ton IA sans vérification humaine.
Qu'en penses-tu ?

Appendix B

Activities Screenshots

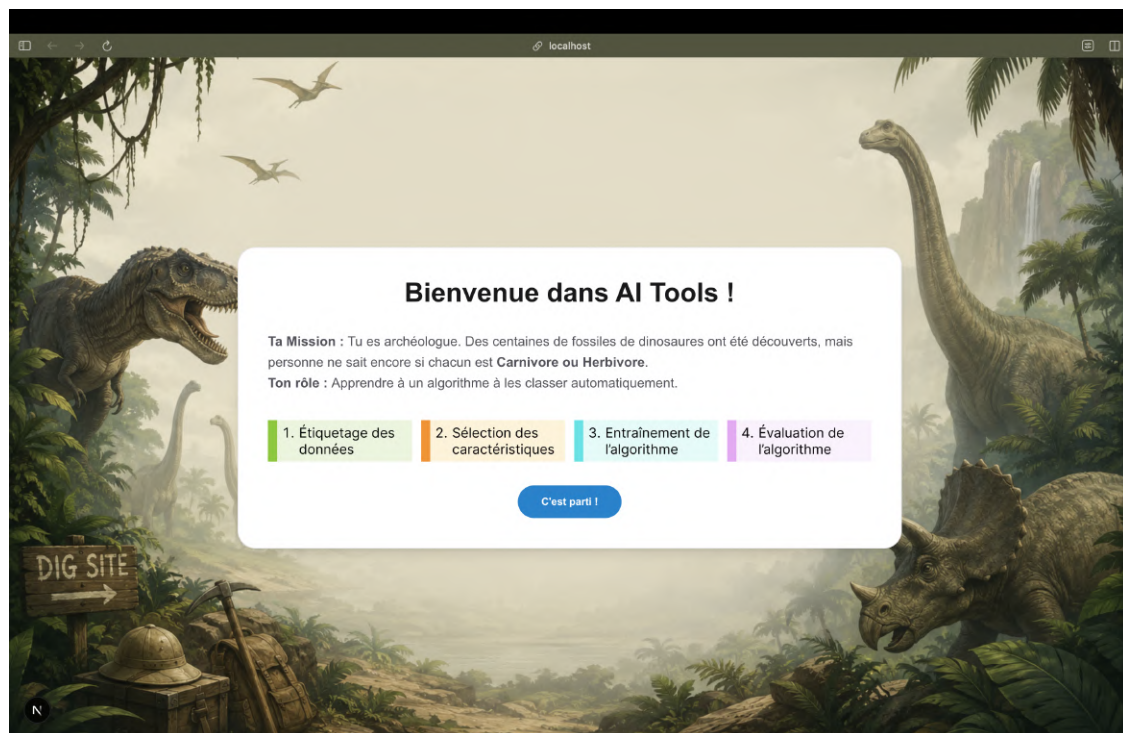
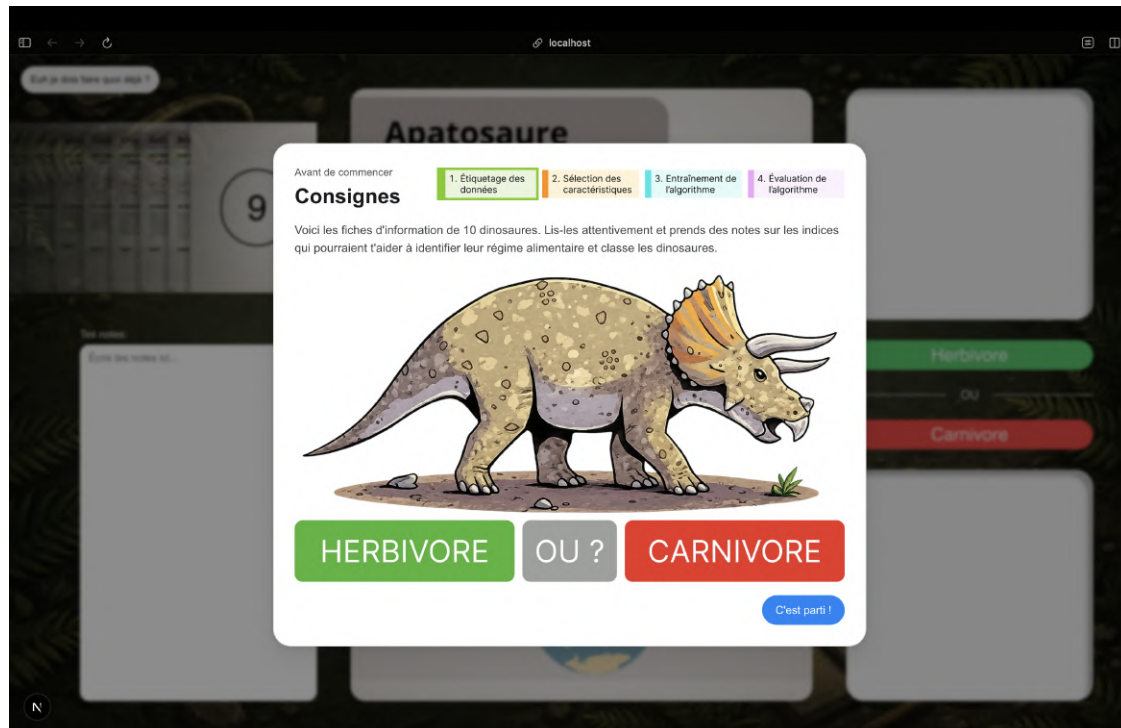


Figure B.1: Welcome word and mission assignment: students will play the role of archaeologists who need to build an AI model to classify dinosaurs between Herbivores and Carnivores

B.1 Data Labelling



(a) Labelling instructions for Glass-Boxed C1 and C2



(b) Labelling instructions for Black-Boxed C3

Figure B.2: Data Labelling instructions for Glass-Boxed conditions (C1 and C2) VS Black-Boxed condition (C3). All groups go through the cards and are asked to read them and take some notes on what could help them identify their diet. The main difference lies in the fact that C1 and C2 need to label the dinosaurs themselves while C3 already has the dinosaurs pre-labelled. (See also Figure 4.1)

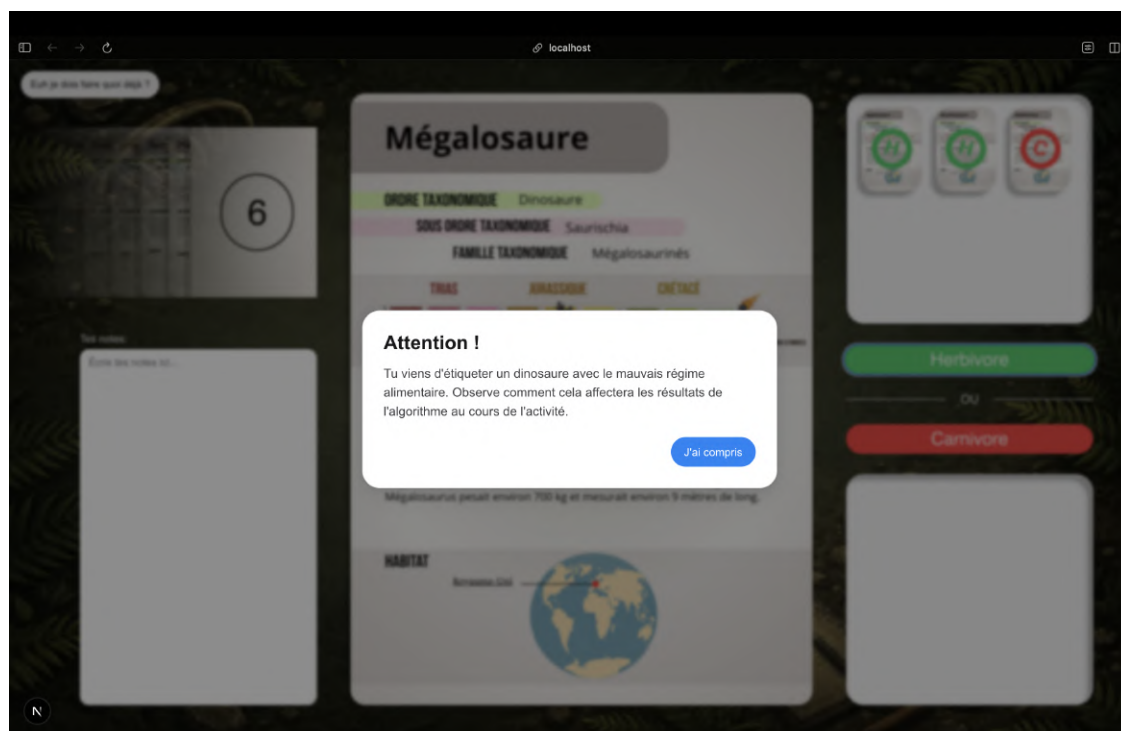


Figure B.3: Warning pop-up when students label a dinosaurs wrongly. (C1 and C2 only)

B.2 Model Training

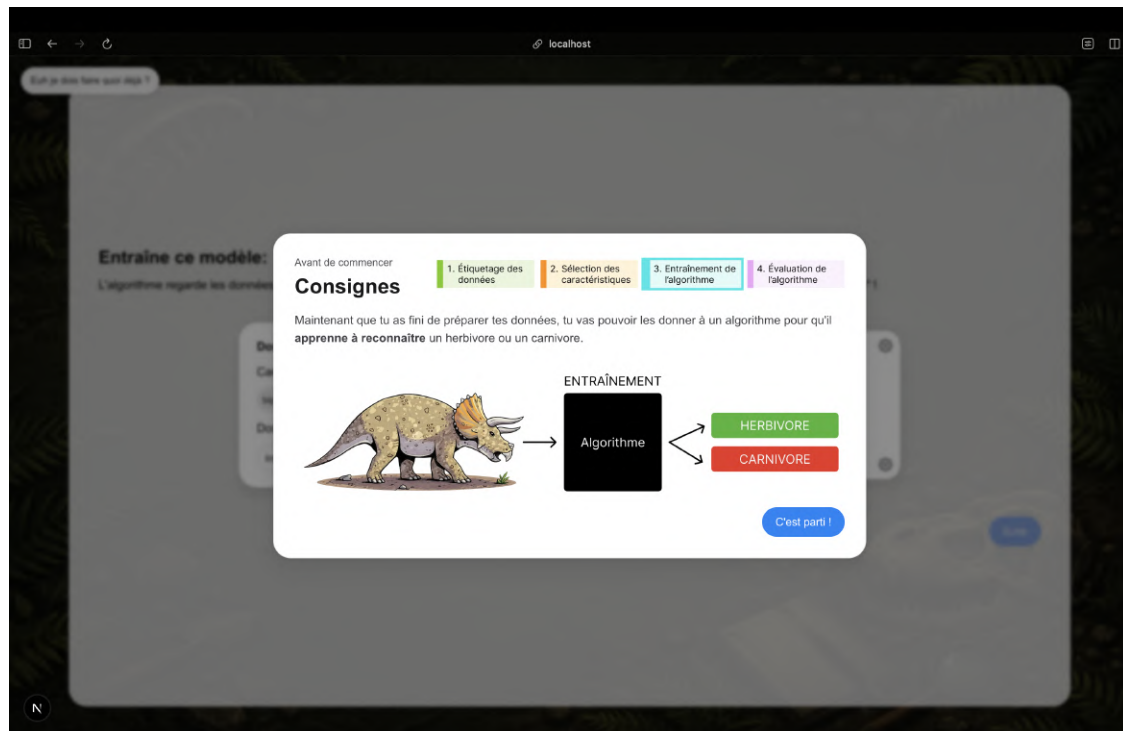


Figure B.4: Model training general instructions for all conditions (C1, C2 and C3).

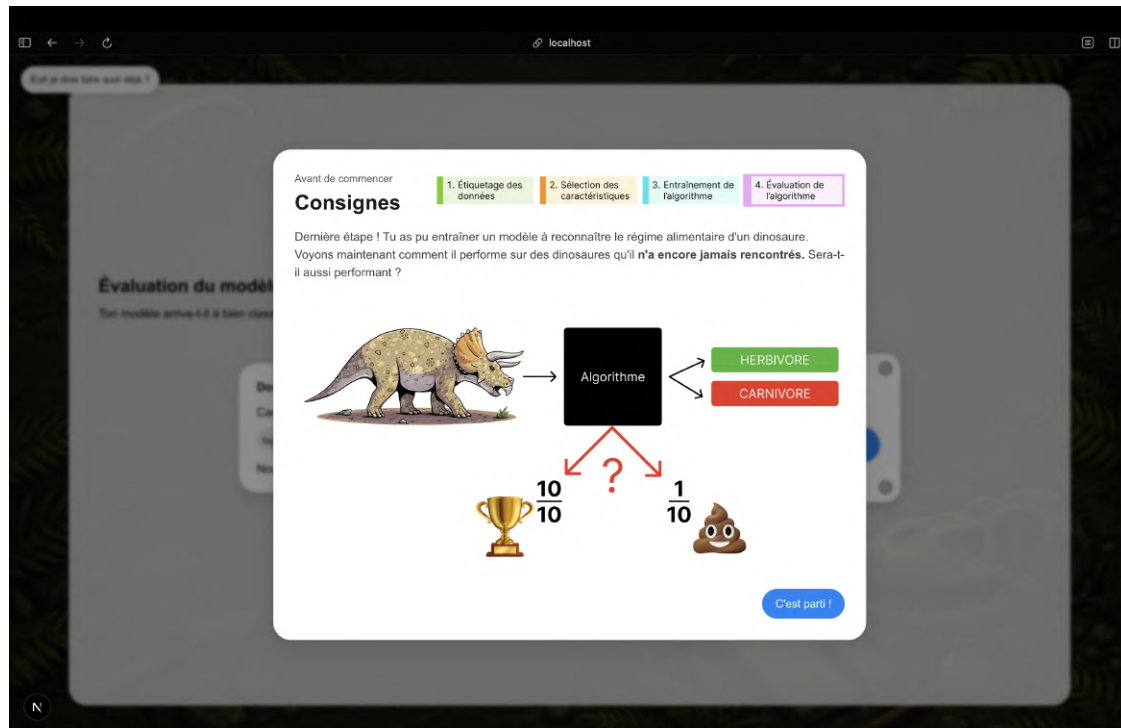


Figure B.5: Evaluation general instructions for all groups (C1, C2 and C3)

B.3 Evaluation

Euh je dois faire quoi déjà ?

Évaluation du modèle

Ton modèle arrive-t-il à bien classer des dinosaures qu'il n'a encore jamais vu ? Calcule sa précision.

Données de test - Modèle

Tri: tableau initial

nom	régime_alimentaire	régime_alimentaire_prédit	bipède	longueur (m)	poids (kg)	famille_taxonomique
Psittacosaurus	herbivore	herbivore	Vrai	2.0	50	Psittacosauridae
Dracorex	herbivore	herbivore	Vrai	4.0	100	Pachycephalosauridae
Pachycephalosaurus	herbivore	herbivore	Vrai	8.0	450	Pachycephalosauridae
Lambeosaurus	herbivore	carnivore	Vrai	9.0	2500	Lambeosaurinae
Atlascopcosaurus	herbivore	herbivore	Vrai	3.0	125	Euornithopoda
Allosaurus	carnivore	carnivore	Vrai	12.0	2000	Allosauridae
Compsognathus	carnivore	herbivore	Vrai	0.65	3	Compsognathidae
Carcharodontosaurus	carnivore	carnivore	Vrai	15.0	6200	Carcharodontosauridae
Mapusaurus	carnivore	carnivore	Vrai	13.0	3000	Giganotosaurinae
Tarbosaurus	carnivore	carnivore	Vrai	10.0	5000	Tyrannosaurinae

Vérifier mes réponses

(a) C1 and C2 see their selected features from the past activities.

Euh je dois faire quoi déjà ?

Évaluation du modèle

Ton modèle arrive-t-il à bien classer des dinosaures qu'il n'a encore jamais vu ? Calcule sa précision.

Données de test - Modèle

Tri: tableau initial

nom	régime_alimentaire	régime_alimentaire_prédit
Psittacosaurus	herbivore	carnivore
Dracorex	herbivore	carnivore
Pachycephalosaurus	herbivore	carnivore
Lambeosaurus	herbivore	herbivore
Atlascopcosaurus	herbivore	carnivore
Allosaurus	carnivore	herbivore
Compsognathus	carnivore	carnivore
Carcharodontosaurus	carnivore	carnivore
Mapusaurus	carnivore	herbivore
Tarbosaurus	carnivore	herbivore

Vérifier mes réponses

(b) C3 can only see the labels.

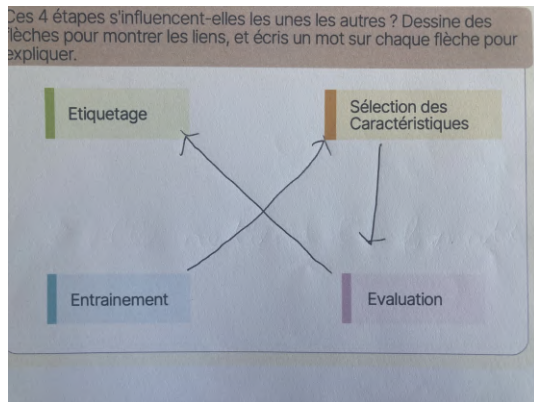
Figure B.6: Prediction tables from the Evaluation activity: students need to compare between true diets and predicted diets to compute its accuracy and assess its performance on unseen data.

Appendix C

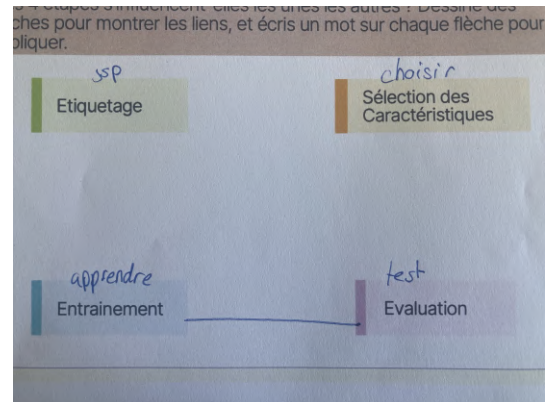
Pilot Experiment Data

Cond.	Group	Pre-test			Post-test					
		Q1: What is AI for you?	Q2: How do you think AI becomes intelligent?	Q3: What happens when an AI learns?	Q1: What is AI for you?	Q2: How do you think AI becomes intelligent?	Q3: What happens when an AI learns?	Q4: Reason for misclassification	Q5: Steps mutual influences	Q6: Museum using AI with no check
C1	1	A software that reflects by itself	It evolves day by day	It retains the information	A software that reflects by itself	It gets better day by day	It retains information	I don't know	IMAGE 1 in Figure C.1a	I say yes and I sell them my AI for 1 BTC
C1	1	A robot that knows how to answer anything	By looking for sources	It transfers the information	Same	By looking for sources	It transfers	Website	IMAGE 2 in Figure C.1b	It is not good because there are personal things
C1	1	A tool to help	By giving it data	I don't know	empty	empty	I don't know	I don't know	IMAGE 3 in Figure C.1c + "Labelling: we make it learn something; Training it is the fact of repeating many times something; evaluation it is when it has learnt and takes the best decisions; selection we choose it to do something"	No because the museum wants that the AI knows everything about ancient objects
C1	2	It is something intelligent inside a screen	It is going to become too intelligent	It searches for the answer because we asked something	I did not change my mind	By working	It is the answers it searches for	A bug	IMAGE 4 in Figure C.1d	No, we need to do tests
C1	2	A tool to help humans	empty	empty	empty	empty	empty	Not enough precision	IMAGE 5 in Figure C.1e	empty
C1	2	A support for everyday	Thanks to the questions that we ask everyday	I don't know	A support for everyday	Did not change my mind	Question, answer	Not enough precision	empty	Ok but I warn them to check
C3	1	It's a software that gives all types of information about anything	Because it is a gigantic database and it develops itself when talking to humans or with websites	I don't know	Same	The house	We give it some information, it analyzes it and it learns	From me	empty	It's not ok
C3	1	A software created to answer your questions, to make some reflections and to create images and texts. It extracts information from databases and the internet.	Thanks to all the databases and the information on the internet	I don't know	Same	Same	We give it info, it learns them and analyzes	From me	IMAGE 8 in Figure C.1f	No
C3	1	It is something that can answer questions and can code websites	Thanks to different researches that it does on Google, Safari and takes different information on different forums	Something	Same	Same	We give it information that it remembers	I don't think that my AI will make mistakes	empty	I would not tolerate this
C3	2	A slave	With humans	It is happy	A slave	With humans	I don't know	From the system	IMAGE 10 in Figure C.1g	It would lose benefits
C3	2	A slave	With humans	It is happy	A slave	With humans	Idk	From the system	IMAGE 11 in Figure C.1h	It would lose benefits

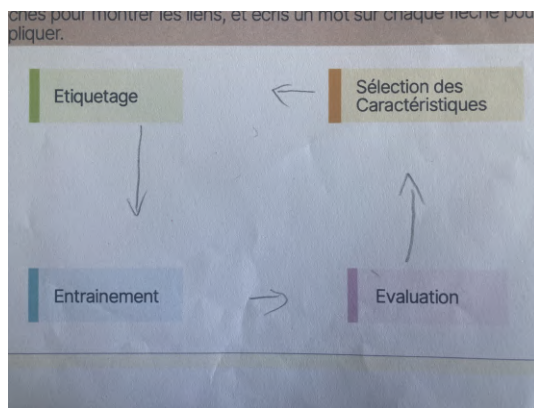
Table C.1: Students' answers to the pre-test and post-test questions.



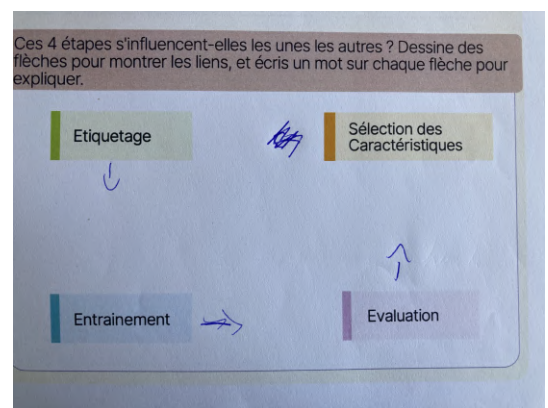
(a) Image 1 (C1 G1)



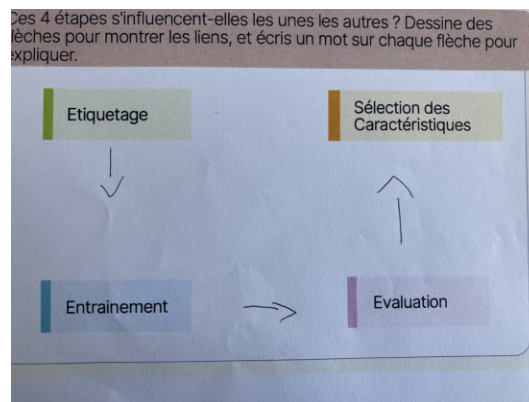
(b) Image 2 (C1 G1)



(c) Image 3 (C1 G1)

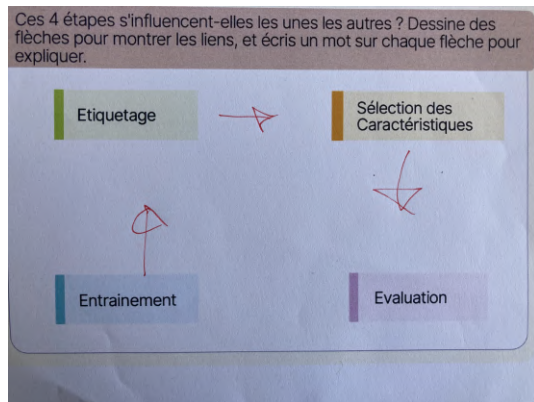


(d) Image 4 (C1 G2)

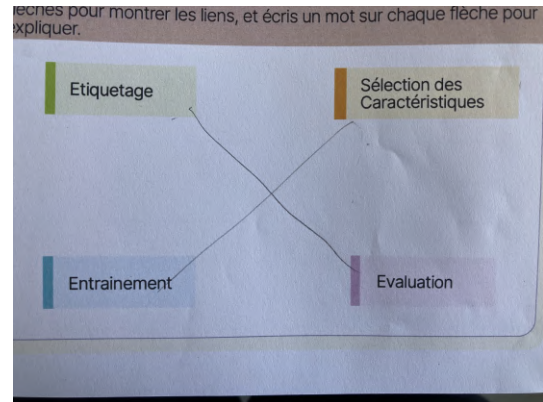


(e) Image 5 (C1 G2)

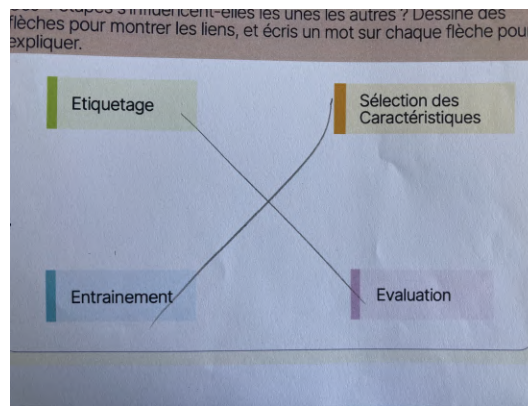
Figure C.1: Answers to Q5 of the Post-Test from students in the C1 condition. Students were asked to draw an arrow to highlight influences between steps and to add a word to describe the influence. One student left the question empty and is not displayed.



(f) Image 8 (C3 G1)



(g) Image 10 (C3 G2)



(h) Image 11 (C3 G2)

Figure C.1: (CONTINUED) Answers to Q5 of the Post-Test from students in the C3 condition. Students were asked to draw an arrow to highlight influences between steps and to add a word to describe the influence. Two students left the question empty and are not displayed.